等离子体定义

参量. indensity 相空间.

温度? 能谱的半高全宽.

热平衡 —— Maxwel 分布.

Debye sheilding 泊松方程.

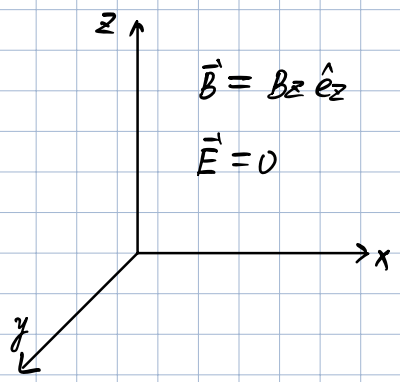
准中性条件.

Single - particle motion

电磁场中 动力学方程:

$$\frac{d(mv)}{dt} = q(\vec{E} + \vec{v}\times\vec{B}) + m\vec{g}$$ 可忽略.

① $\vec{E} = 0$



$\vec{B} = B_z \hat{e}_z$

$\vec{E} = 0$

$$\begin{cases} \dot{v}_x = -\omega_c v_y \\ \dot{v}_y = \omega_c v_x \end{cases} \qquad \ddot{v}_x = -\omega_c \dot{v}_y = -\omega_c^2 v_x$$

$$\downarrow$$

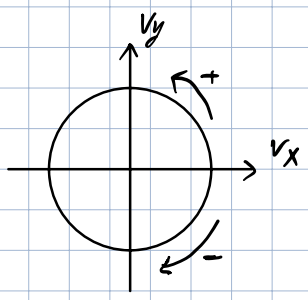
$$\begin{cases} \ddot{v}_x = -\omega_c^2 v_x \\ \ddot{v}_y = -\omega_c^2 v_y \end{cases} \Rightarrow \quad v_{x,y} = v_\perp e^{\pm i\omega_c t + \delta_{x,y}}.$$

令 对 v_x, $\delta_x = 0$. $\quad v_x = v_\perp e^{i\omega_c t}$,

$$v_y = -\frac{1}{\omega_c}\dot{v}_x = \pm\underset{u}{i} v_\perp e^{i\omega_c t} = \pm v_\perp e^{i(\omega_c t + \frac{\pi}{2})}.$$

note: ω_c : Lamor frequency $\quad \omega_c = \left|\frac{qB}{m}\right|$

Q: ± 是如何出现的?



$$\dot{v}_x = v_\perp e^{i\omega_c t}, \quad \dot{v}_y = \pm v_\perp e^{i(\omega_c t + \frac{\pi}{2})}, \quad (\pm : q)$$

$$x - x_0 = -i\frac{v_\perp}{\omega_c} e^{i\omega_c t} = r_L \sin\omega_c t$$

$$y - y_0 = \pm\frac{v_\perp}{\omega_c} e^{i\omega_c t} = \pm r_L \cos\omega_c t$$

Def. Lamor radii $\quad r_L = \frac{v_\perp}{\omega_c} = \frac{mv_\perp}{|q|B}$

$$(x-x_0)^2 + (y-y_0)^2 = r_L^2$$ 实际就是1个圆.

considering electrical field

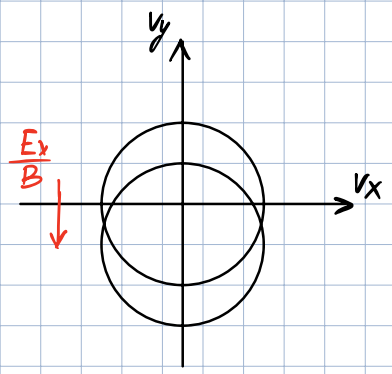
$$\frac{d(m\vec{v})}{dt} = q(\vec{E} + \vec{v}\times\vec{B}).$$

$$\begin{cases} \frac{dv_z}{dt} = \frac{q}{m}E_z. \Rightarrow v_z = \frac{q}{m}E_z t + v_{z,0}. \\ \frac{dv_x}{dt} = \frac{q}{m}E_x \pm \omega_c v_y, \\ \frac{dv_y}{dt} = \mp\omega_c v_x. \end{cases}$$

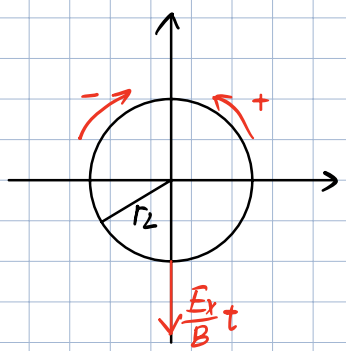
$$\begin{cases} \ddot{v}_x = \pm\omega_c \dot{v}_y = \omega_c^2 v_x. \\ \ddot{v}_y = \mp\omega_c \dot{v}_x = \mp\frac{\omega_c q}{m}E_x + \omega_c^2 v_y \end{cases}$$

$$\begin{cases} \ddot{v_x} - \omega_c^2 v_x = 0 & \Rightarrow \quad v_x = v_\perp e^{i\omega_c t} \\ \ddot{v_y} - \omega_c^2 v_y \pm \frac{\omega_c B}{m} E_x = 0 & \Rightarrow \quad v_y = \pm i v_\perp e^{i\omega_c t} - \frac{E_x}{B}. \end{cases}$$

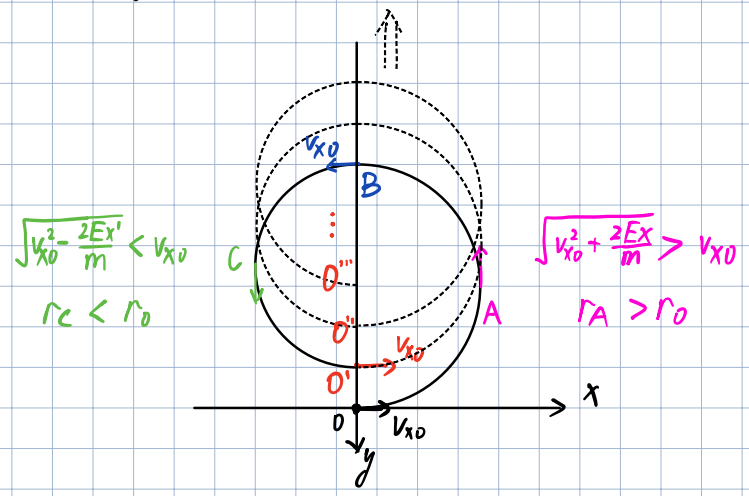
相当于在无电场的结果上加上一沿 y 方向的平动.



$$\begin{cases} x - x_0 = -i\frac{v_\perp}{\omega_c} e^{i\omega_c t} = r_L \sin\omega_c t \\ y - y_0 = \pm\frac{v_\perp}{\omega_c} e^{i\omega_c t} = \pm r_L \cos\omega_c t - \frac{E_x}{B} t. \end{cases}$$



Q: 无外电场下，粒子沿磁场圆周运动. 当沿x方向施加电场$\vec{E}$，为何粒子反而沿y运动？



曲率半径 $r(O\to B)>r_0>r(B\to O)$.

每个周期内前半周与后半周曲率半径不同.

$$m\frac{dv}{dt}=q(\vec{E}+v\times\vec{B})=0.$$

$$\vec{E}+\vec{v}\times\vec{B}=0 \qquad \vec{B}\times\vec{E}+\vec{B}\times(\vec{v}\times\vec{B})=0.$$

$$\vec{E}\times\vec{B}=\vec{B}\times(\vec{v}\times\vec{B})=\vec{v}B^2-\vec{B}(\vec{v}\cdot\vec{B})$$

$$v_\perp=\frac{\vec{E}\times\vec{B}}{B^2}=v_E.$$ 电漂移.

对一般力场，

$$v_\perp=\frac{\vec{F}\times\vec{B}}{qB^2}$$

考虑重力场 mg，

$$\vec{v}_g = \frac{m\vec{g}\times\vec{B}}{qB^2}$$

不同之处，离子与电子向相反方向漂移.

∴ 等离子中存在净电流密度.

$$\vec{j} = n(M+m)\frac{\vec{g}\times\vec{B}}{B^2}.$$